VITALITY 08

Alprostadil

Drug form of prostaglandin E1, a natural fatty-acid messenger; for ED, injected into the penis



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Prostaglandin	Erectile dysfunction	Large trials; FDA approved	FDA approved (Caverject, Edex)

STUDIED FOR

Erections (injection)	Erections (urethral)	Newborn heart defects
HUMAN	HUMAN	HUMAN

Trimix is not FDA approved. Compounded Trimix mixes alprostadil with papaverine and phentolamine. In one trial it caused more priapism than alprostadil alone.

HOW IT WORKS IN THE BODY

Alprostadil ▶ PGE1 receptors ▶ More cAMP ▶ Blood trapped

- **Binds its receptors** in the penis and raises cAMP (label)
- **Relaxes spongy smooth muscle** and widens penile arteries
- **Filled spaces press** the small veins shut, trapping blood

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved for ED: Caverject 1995, Edex 1997
- Self-injection study: 683 men, 6 months
- Urethral trial (Muse): 1,511 men

NOT YET ESTABLISHED

- Combining with other injections
- Bioavailability (penile route)

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
Not established for penile injection	Injection into the penis; urethral pellet (now discontinued)
PHARMACOKINETICS	**SOLD AS**
Main metabolite peaks at 30 minutes, back to baseline by 60	Prescription injection (Caverject, Caverject Impulse, Edex)
MONITORED IN TRIALS	**SPORT (WADA)**
Penile exam for scarring, erections over 4 hours, blood pressure	Not on the 2026 list

REGULATORY

FDA	Approved for ED: Caverject 1995, Edex 1997
FDA	Compounded Trimix is not approved
EMA	No EU-wide authorisation; national approvals
WADA	Not on the 2026 list

REFERENCES

1. FDA, Caverject prescribing information (DailyMed), May 2024
2. Linet and Ogrinc, N Engl J Med, intracavernosal alprostadil, Apr 1996
3. Padma-Nathan et al., N Engl J Med, urethral alprostadil (Muse), Jan 1997

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